Campus Events – Real API Integration and Failure Exercise

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Device: Android emulator `ex1` ("Android SDK built for x86_64"), Android 13 (API 33), 1080x2400. Cold-booted with `-dns-server 8.8.8.8`.

App: `com.inci.campuscompanion`, debug build

GitHub PR link: https://github.com/toufic-fy/Campus-Companion/pull/3

Every result below was observed on the emulator through `adb`: UI hierarchy dumps (`uiautomator`), screenshots (`adb exec-out screencap -p`, each one opened and checked), and `adb logcat`.

Changes after teacher review

- **API source.** The app now fetches events from the API posted on Moodle: `GET https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/events/event`. Before coding, `curl` returned HTTP 200 and a JSON array of 50 objects, each with 8 string fields.
- **`CampusApi`.** `BASE_URL = "https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/"` and `@GET("events/event") suspend fun getEvents(): List<EventDto>`.
- **`EventDto`.** Rewritten to mirror the JSON field for field: `id`, `createdAt`, `name`, `avatar`, `startingDate`, `endingDate`, `description`, `address`. All are `String?`. The names match the JSON exactly, so no `@SerializedName` is needed.
- **`CampusEvent.id`.** Changed from `Int` to `String`, because the API returns `"id": "1"`.
- **`EventDto.toDomain()`** in `EventsRepository.kt`:
  - `title` comes from `name`.
  - `location` comes from `address`, with fallback `"TBA"`.
  - `startsAt` comes from `startingDate`, formatted by `formatEventDate()`.
  - `description` is kept.
  - Items with a blank `id` or `name` are still dropped.
- **Date parsing.** minSdk is 24, and the project has no core-library desugaring, so `java.time` is not used. `formatEventDate()` parses the ISO-8601 UTC string with `SimpleDateFormat` (pattern `yyyy-MM-dd'T'HH:mm:ss.SSS'Z'`, then the same without millis), with `isLenient = false`. It formats as `MMM d, yyyy · HH:mm` in the device time zone. If parsing fails, it returns the raw string, so bad data cannot crash the list.
- **Old data source removed.** `api/events.json` was deleted from the project folder. `grep -rn "raw.githubusercontent\|events.json" app/src` finds nothing.
- **Unchanged.** HTTPS, the `INTERNET` permission, the Gson converter, Loading / Content / Empty / Error + Retry, `toUserMessage()`, the `Log.w` in the repository, and the `CancellationException` rethrow.

Real API data on screen (network on, first launch). The first item, `"startingDate":"2026-10-20T12:23:40.729Z"`, is shown as `Oct 20, 2026 · 15:23` in the emulator's UTC+3 time zone:



Summary

#	Scenario	Expected	Observed	Matched on first try	Fix needed
1	Network off (airplane mode)	Error, no crash	Error + Retry; Retry gives Error again; same PID	Yes	No
2	Invalid base URL + Retry	Retryable Error	Error + Retry; each Retry logs a new `UnknownHostException`	Yes	No
3	Endpoint returns `[]`	Dedicated Empty message	"No upcoming events.", no Error title, no Retry button	Yes	No
4	Restore endpoint, network on, Retry	Content returns	Real API list shown after Retry	Yes	No

Extras (temporary mock): HTTP 404, HTTP 500 and an object-instead-of-array body each mapped to Error with the correct category message. No crash.

Scenario 1 – Disconnect the network

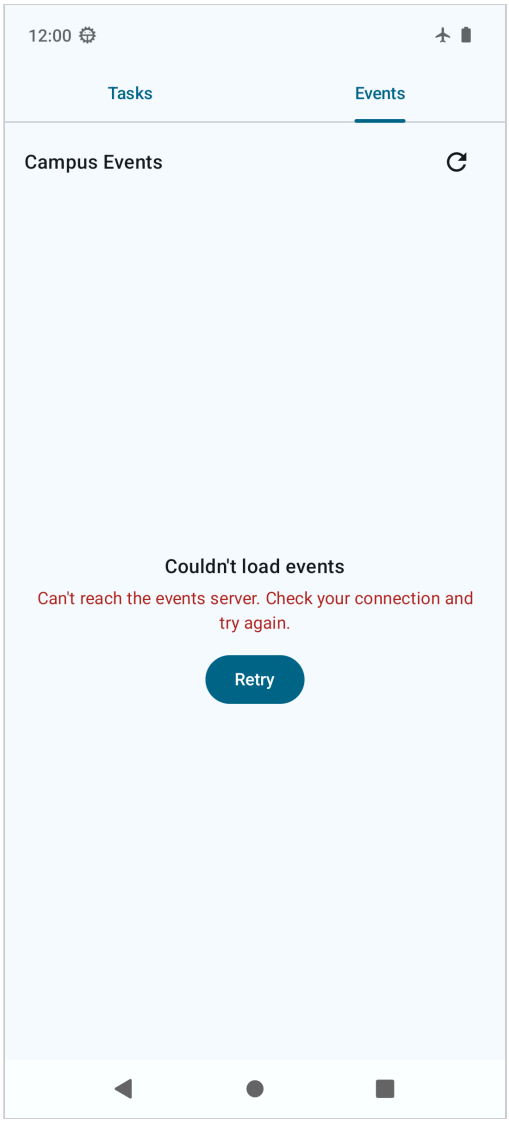
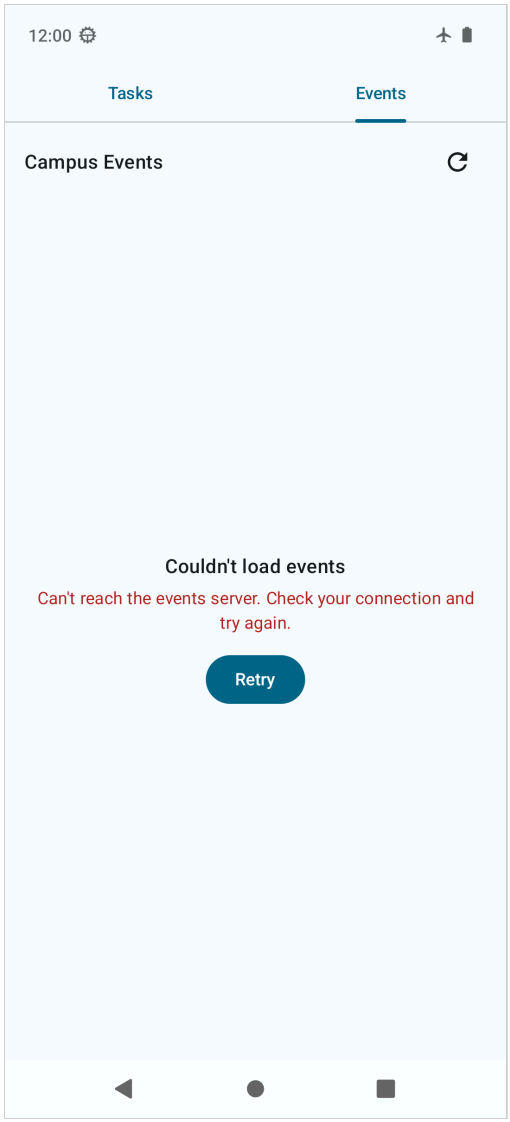
Change: no code change. Network turned off with `adb shell cmd connectivity airplane-mode enable`. `dumpsys connectivity` reported `Active default network: none`. The app was force-stopped, launched, and the Events tab opened. Then Retry was pressed.

Observed:

- On launch, the Error state: "Couldn't load events" / "Can't reach the events server. Check your connection and try again." / Retry.
- After Retry, the same Error state.
- The process ID stayed `2245` before and after Retry. No crash, and no `FATAL` line in logcat.

Logcat:

```
10-27 12:00:19.733  2245  2245 W EventsRepository: Failed to load events: java.net.UnknownHostException: Unab
le to resolve host "6ad70838f71f154ac4879fac.mockapi.io": No address associated with hostname
10-27 12:00:32.683  2245  2245 W EventsRepository: Failed to load events: java.net.UnknownHostException: Unab
le to resolve host "6ad70838f71f154ac4879fac.mockapi.io": No address associated with hostname
```

Launch (offline)	After Retry (offline)
	

Matched expected behavior: Yes. **Fix:** none needed.

Scenario 2 – Invalid base URL

Change (temporary, reverted): in `app/src/main/java/com/inci/campuscompanion/data/remote/CampusApi.kt`, line 12:

```
- private const val BASE_URL = "https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/"
+ private const val BASE_URL = "https://campus-events.invalid/api/v1/"
```

Network was on (`Active default network: 103`, and the real API host resolved from the emulator shell). The app was rebuilt, installed and launched, then Retry was pressed.

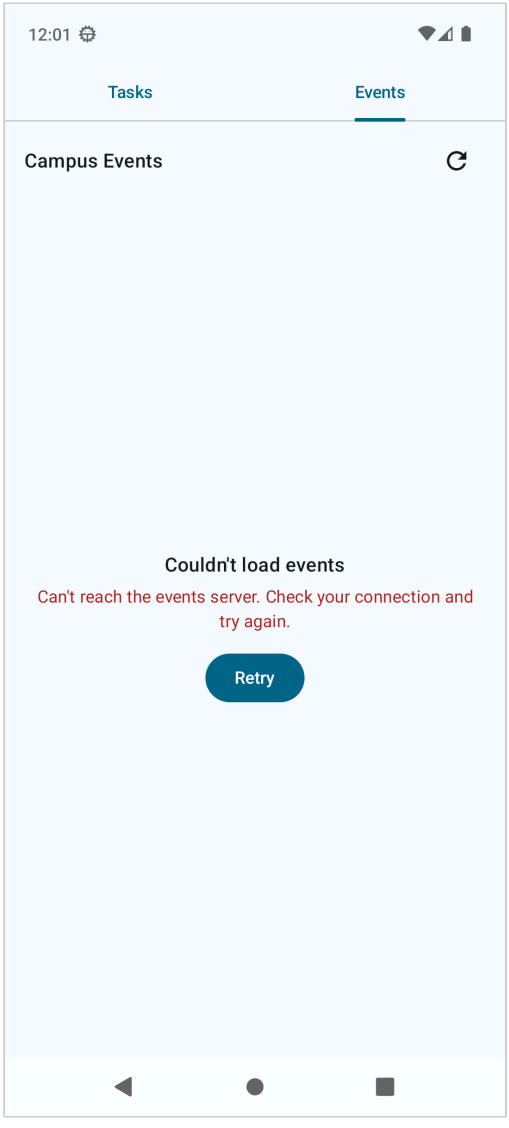
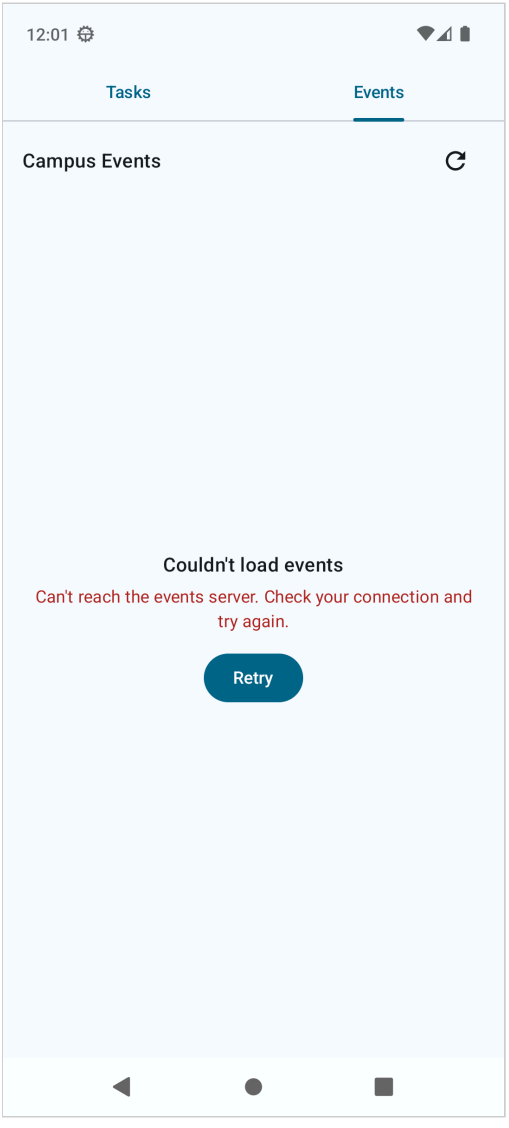
Observed:

- Error state with "Can't reach the events server. Check your connection and try again." and Retry, on launch and again after Retry.
- Each Retry sent a new request (one log line per attempt).

- PID `2479`, no crash.

Logcat:

```
10-27 12:01:07.515  2479  2479 W EventsRepository: Failed to load events: java.net.UnknownHostException: Unab
le to resolve host "campus-events.invalid": No address associated with hostname
10-27 12:01:20.447  2479  2479 W EventsRepository: Failed to load events: java.net.UnknownHostException: Unab
le to resolve host "campus-events.invalid": No address associated with hostname
```

Launch	After Retry
	

Matched expected behavior: Yes. **Fix:** none needed.

Scenario 3 – Empty JSON array

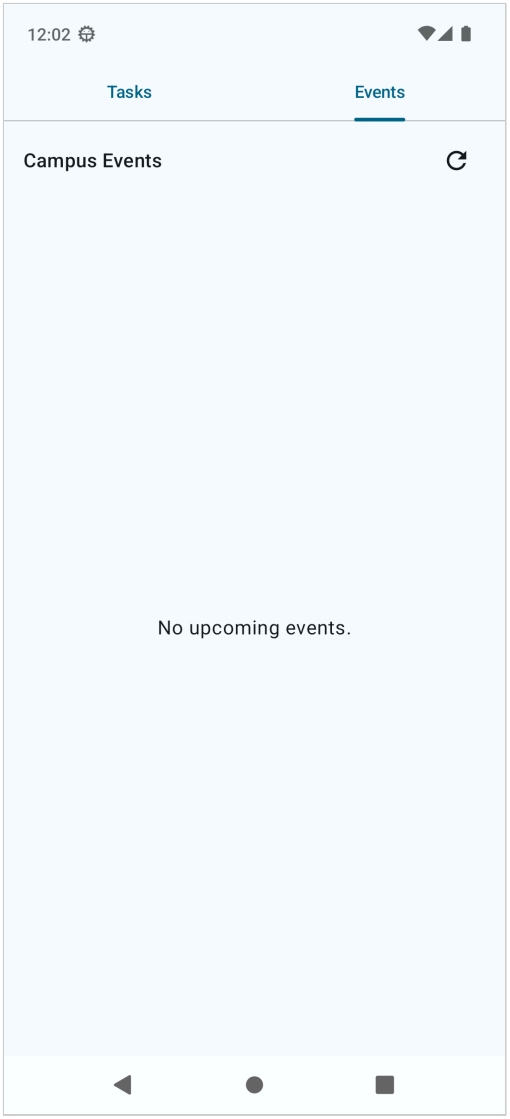
Change (temporary, reverted): `CampusApi.kt` was first restored to the valid URL. Then a temporary OkHttp client was added to `Retrofit.Builder()`. Its interceptor answers every request with HTTP 200 and body `[]`, so the response still goes through Retrofit, the Gson converter, `EventsRepository` and `EventsViewModel`:

```
.client(OkHttpClient.Builder().addInterceptor { chain ->
    Response.Builder().request(chain.request()).protocol(Protocol.HTTP_1_1)
        .code(MOCK_CODE).message("mock")
        .body(ResponseBody.create(MediaType.get("application/json"), MOCK_BODY))
        .build()
}.build())
```

with `MOCK_CODE = 200` and `MOCK_BODY = "[]"`.

Observed:

- Only "No upcoming events." is shown. There is no "Couldn't load events" title and no Retry button, so it is clearly different from Error.
- No `EventsRepository` line in logcat, because this is the success path that leads to `EventsUiState.Empty`.
- PID `2648`, no crash.

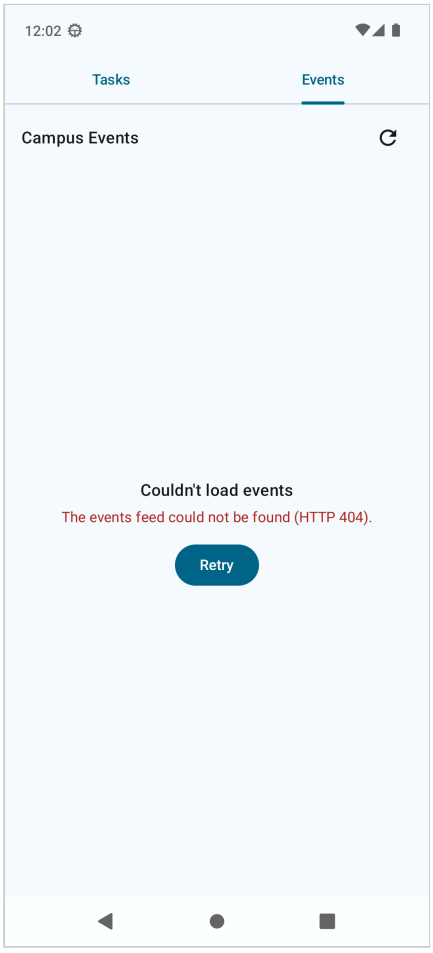
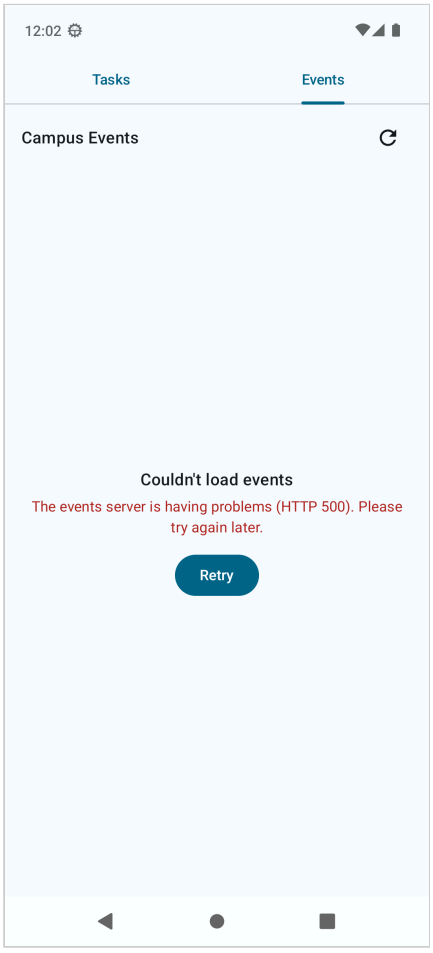
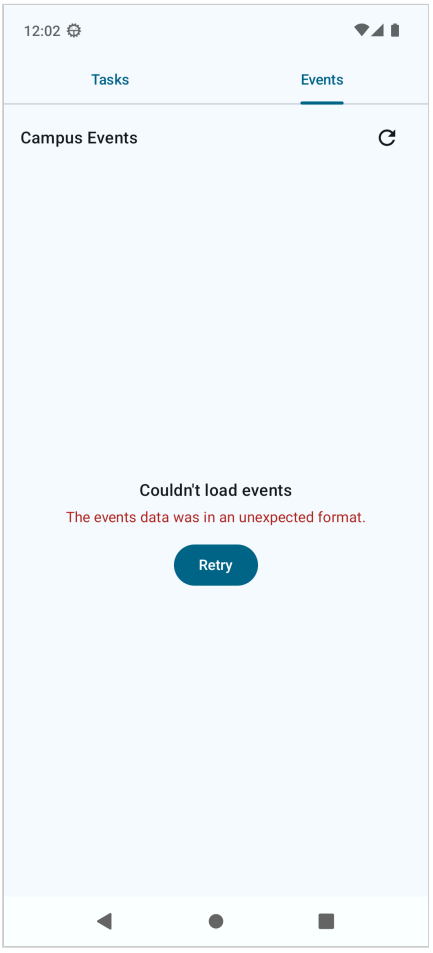


Matched expected behavior: Yes. **Fix:** none needed.

Extras – HTTP 404, HTTP 500, malformed body

Same temporary interceptor, with only `MOCK_CODE` / `MOCK_BODY` changed for each run. The interceptor was removed afterwards.

Mock response	UI message (Error + Retry)	Logcat
`404 {"error":"not found"}`	The events feed could not be found (HTTP 404).	`W EventsRepository: Failed to load events: retrofit2.HttpException: HTTP 404 mock`
`500 {"error":"boom"}`	The events server is having problems (HTTP 500). Please try again later.	`W EventsRepository: Failed to load events: retrofit2.HttpException: HTTP 500 mock`
`200 {"events":[]}`	The events data was in an unexpected format.	`W EventsRepository: Failed to load events: java.lang.IllegalStateException: Expected BEGIN_ARRAY but was BEGIN_OBJECT at line 1 column 2 path $`

HTTP 404	HTTP 500	Object instead of array
		

Scenario 4 – Restore the valid endpoint and Retry

Change:

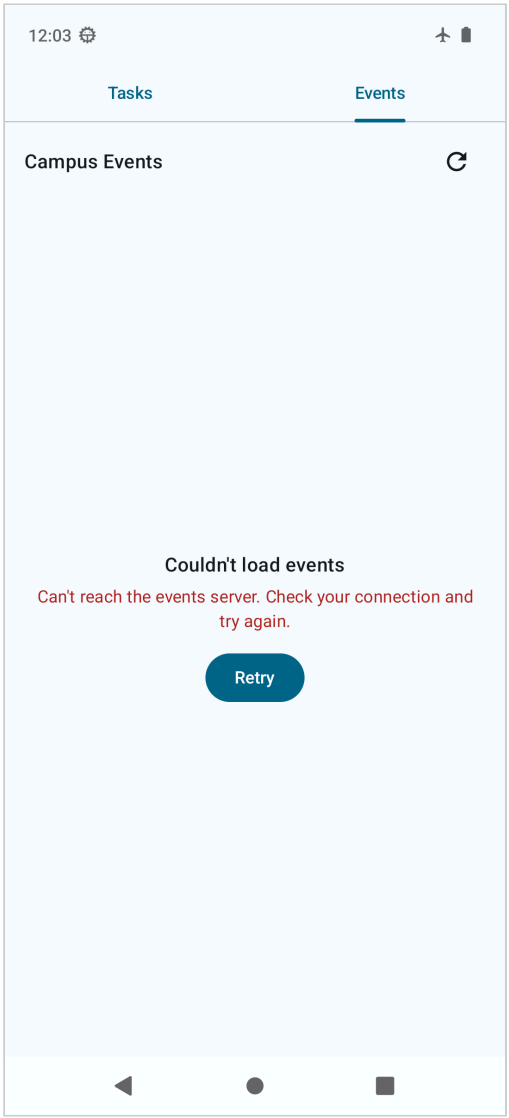
- `CampusApi.kt` was restored from the backup taken before any temporary change. `cmp` reported the file byte-for-byte identical.
- `grep -rn "TEMP\|MOCK_\|\.invalid\|okhttp3" app/src` found nothing.
- The app was rebuilt and installed.

Steps:

1. Network off (`airplane-mode enable`), app launched. Error state shown.
2. Network on (`airplane-mode disable`). The API host resolved from the emulator about 6 s later, and `Active default network: 104` was reported.
3. Retry pressed in the same running app.

Observed:

- The real API list returned: "Ms. Elvira Homenick · Oct 20, 2026 · 15:23 · 819 The Laurels", "Shane Zulauf-Zulauf", "Heather Kuvalis", "Sharon Wisozk", …
- No `EventsRepository` failure line in logcat after Retry.
- PID `3102`, no crash.

Before Retry (offline)	After Retry (online)
	

Matched expected behavior: Yes. **Fix:** none needed.

Failure classification

The mapping is done by `EventsViewModel.toUserMessage()`. The repository rethrows `CancellationException`, so it never reaches this mapping.

Category	Exception (as observed / handled)	Message
No connection / bad host	`IOException`, e.g. `UnknownHostException`	Can't reach the events server. Check your connection and try again.
Timeout	`SocketTimeoutException`	The server took too long to respond. Please try again.
HTTP 4xx	`HttpException` 400–499	The events feed could not be found (HTTP xxx).
HTTP 5xx	`HttpException` 500–599	The events server is having problems (HTTP xxx). Please try again later.
Serialization / data mismatch	`IllegalStateException`, `JsonParseException`, `MalformedJsonException`, `EOFException`	The events data was in an unexpected format.

A success with an empty list becomes `Empty`, not `Error`.

Layers explanation – one refresh action

Request path (down):

- **Compose UI.** `EventsScreen(uiState, onRefresh)`
  - `MainActivity` collects `eventsViewModel.uiState` with `collectAsStateWithLifecycle()` and passes it in.
  - Tapping **Retry** (`EventsError`) or the refresh `IconButton` calls `onRefresh`, which is `eventsViewModel.refresh()`.
  - The UI only renders state. It does no loading itself.
- **ViewModel.** `EventsViewModel.refresh()`
  - Sets `_uiState.value = EventsUiState.Loading`, so the UI shows `CircularProgressIndicator`.
  - Launches a coroutine in `viewModelScope` and calls `eventsRepository.getEvents()`.
- **Repository.** `EventsRepository.getEvents(): Result<List<CampusEvent>>`
  - Calls `api.getEvents()` directly on the injected `CampusApi`. There is no separate remote-data-source class; `CampusApi` is the remote layer.
  - Rethrows `CancellationException`. Any other `Exception` is logged with `Log.w("EventsRepository", ...)` and returned as `Result.failure(e)`.
- **Retrofit service.** `CampusApi` (created once in `MainActivity` by `CampusApi.create()`)
  - `@GET("events/event") suspend fun getEvents(): List<EventDto>` with `BASE_URL = "https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/"`.
  - Retrofit builds the HTTPS request; `GsonConverterFactory` is registered for the body.
- **Network.** OkHttp (Retrofit's default client) sends `GET https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/events/event`.

Response path (up):

- **Network / Retrofit result.**
  - `2xx`: Gson parses the JSON array into `List<EventDto>`, with field names matching the JSON.
  - Non-`2xx`: Retrofit throws `HttpException`.

    - No connection or unknown host: OkHttp throws an `IOException` (`UnknownHostException`).
    - Wrong JSON shape: Gson throws `IllegalStateException` / `JsonSyntaxException`.
- **DTO → CampusEvent mapping** (`EventsRepository`, `mapNotNull { it.toDomain() }`)
    - Items with a blank `id` or `name` are dropped.
    - `CampusEvent(id, title = name, location = address ?: "TBA", startsAt = formatEventDate(startingDate) ?: "Time TBA", description = description.orEmpty())`
    - `formatEventDate()` falls back to the raw string if parsing fails.
- **State update** (`EventsViewModel`, `Result.fold`)
    - Success with a non-empty list: `EventsUiState.Content(events)`. `EventList` renders `EventRow` cards.
    - Success with an empty list: `EventsUiState.Empty`. Shows "No upcoming events.".
    - Failure: `EventsUiState.Error(throwable.toUserMessage())`. `EventsError` shows the message and a Retry button.
    - The new `StateFlow` value triggers recomposition of `EventsScreen`.

Restored configuration

- `BASE_URL`: `https://6ad70838f71f154ac4879fac.mockapi.io/api/v1/`
- Endpoint: `@GET("events/event")`
- `CampusApi.kt` is byte-for-byte identical to the pre-exercise backup. It has no interceptor, no `.invalid` host, and no OkHttp imports.
- `api/events.json` is deleted. No reference to `raw.githubusercontent.com` remains in `app/src`.
- Emulator: airplane mode `disabled`, network on, DNS resolving the API host.
- Final state: the installed build shows **Content** with real API data.


12:03
Tasks
Events
Campus Events
Ms. Elvira Homenick
Oct 20, 2026 · 15:23 · 819 The Laurels
Cribro nostrum dolor absum perferendis demulceo. Debitis porro defendo alii volutabrum degusto vespillo tactus sonitus ustulo. Textor cometes solio quisquam altus cernuus facilis voveo venia velit.
Shane Zulauf-Zulauf
Oct 20, 2026 · 22:55 · 3240 Beer Keys
Tepesco dicta suus amitto. Carpo charisma tutis defaeco vilitas cerno quam.
Heather Kuvalis
Oct 20, 2026 · 23:56 · 9645 Stoltenberg Glen
Sui illum cum aggredior debilito repudiandae. Delectus tibi taceo explicabo deinde adnuo. Centum deprimo deludo ex volup et.
Sharon Wisozk
Oct 21, 2026 · 00:27 · 81071 Guadalupe Lane
Repellat cotidie tres umbra nisi carbo alveus sustineo video assumenda. Ventus assumenda vir cupressus corpus amaritudo auxilium. Distinctio depromo cedo crastinus. Adipiscor voluntarius adipisci volo audio defleo tutamen utrum. Coadunatio velociter basium.
Christina Daugherty